

TASK 14: LINUX SERVER HARDENING & SECURE CONFIGURATION (DETAILED TECHNICAL REPORT)

1. Executive Summary

This project focuses on implementing structured Linux server hardening techniques aligned with industry security standards. The server environment was analyzed for potential vulnerabilities and systematically configured to minimize exposure. Security controls were applied at authentication, network, service, file-system, and update-management levels. The objective was to reduce attack surface, prevent unauthorized access, and ensure system resilience against common cyber threats.

2. Threat Model Consideration

Potential threats considered during hardening include brute-force attacks, privilege escalation, unauthorized remote access, misconfigured services, outdated software vulnerabilities, and weak authentication mechanisms. By identifying these risks, preventive configurations were implemented to mitigate exploitation opportunities.

3. SSH Security Hardening

Root login was disabled by modifying the SSH daemon configuration file (`/etc/ssh/sshd_config`). The parameter `'PermitRootLogin no'` was enforced to prevent direct administrative access via SSH. This significantly reduces the risk of automated brute-force attacks targeting privileged accounts. The SSH service was restarted to apply secure configuration changes.

4. Password Policy Enforcement

Strong password policies were implemented using the `libpam-pwquality` module. Minimum password length, uppercase, lowercase, numeric, and special character requirements were enforced. This ensures resilience against dictionary attacks and password-guessing techniques.

5. Firewall Configuration (UFW)

The Uncomplicated Firewall (UFW) was installed and enabled to control inbound and outbound traffic. Only essential ports such as 22 (SSH) and 80 (HTTP) were allowed. All other unused ports were denied by default. Firewall rules were verified to confirm active protection against unauthorized network access.

6. Service Minimization

Active services were enumerated using `systemctl` commands. Unnecessary services were disabled and stopped to reduce system exposure. Service minimization reduces potential exploitation points for attackers targeting vulnerable services.

7. File Permission Hardening

Sensitive system files such as `/etc/shadow` were inspected to ensure restrictive permissions. Permissions were adjusted using `chmod 600` where required to ensure only the root user has read/write access. This protects critical authentication data from unauthorized modification.

8. Automatic Security Updates

The `unattended-upgrades` package was installed and configured to enable automatic security patching. This ensures timely mitigation of newly discovered vulnerabilities and maintains system integrity.

9. Log Monitoring & Incident Awareness

Authentication logs located at `/var/log/auth.log` were continuously monitored using `tail` commands. Suspicious login attempts, failed authentication events, and abnormal SSH activity were observed and documented. Log monitoring enhances visibility into potential security incidents and supports proactive defense mechanisms.

10. Risk Mitigation Impact Analysis

The implemented controls collectively reduce system attack surface, limit privilege misuse, prevent unauthorized remote access, enforce authentication strength, and ensure ongoing vulnerability remediation. These configurations align with baseline security hardening standards followed in enterprise environments.

11. Conclusion

Through systematic hardening measures, the Linux server environment was significantly strengthened against common threat vectors. The project demonstrates practical application of system security principles including access control, network filtering, service restriction, file integrity protection, and log-based monitoring.

12. Recommended Future Enhancements

Future security improvements may include deploying Fail2Ban for automated brute-force blocking, implementing an Intrusion Detection System (IDS), integrating centralized SIEM solutions for log correlation, and conducting periodic vulnerability scanning and penetration testing.